

Applications of AI in Negotiation

1 Introduction

With the onset of the AI revolution, it's possible that mediators could have access to vastly more data about the parties they're working with, leading to more informed dialogues and greater positive outcomes. Negotiators can use machine learning (ML) to make better predictions about parties' interests when negotiating. However, the limitations of these models bring into question how much of a benefit they truly offer.

2 Informing Strategy

There are numerous applications of AI at the strategy ideation stage of negotiation, which may provide mediators with better information or predictions of possible outcomes. With expedited data processing, these tools can identify key anomalies which may lead to conflict (Choyal et al., 2024). By considering data from social media and news, as well as potential economic indicators, early signs of violence are less likely to go unnoticed. This may allow negotiators to practice a sort of preemption. If they believe it necessary in a certain case, they may bring parties to a table before a conflict breaks out in order to prevent its start in the first place.

Historical strategies can be useful in informing present ones due to the benefit of hindsight. Being able to comprehend what did or didn't work prior allows those involved to revisit former areas of agreement or target areas that are still weak (Choyal et al., 2024).

This necessity can be seen in the work of Arana-Catania et al., who have been collaborating with the Yemeni mediation team to build ML models informed by prior discussions between the parties involved. They predetermined categories and query words which their model could then use to target dialogue about those specific ideas which, combined with a latent issue extraction tool, provided mediators with a full understanding of how these parties have developed. This highlights to them the most contentious issues, helping them focus their time. It also helped determine the latent issues which could lead to more consensuses.

3 During Negotiation

Most of the benefits of ML to the active negotiation process stem from the natural language processing (NLP) capabilities of these models. In negotiation, the mediator works to understand to the best of their ability what the parties are thinking. They use a combination of verbal and non-verbal (proxemic) cues to make educated guesses about how they may be reacting to the issue at hand. By employing NLP, the context of the negotiation can be interpreted by the model, providing it a strong basis to glean reactive cues. Combining this method with sentiment analysis helps identify points of contention as well as points of agreement in order to fully understand the subtext of what's being said.

Negotiations often occur between parties that have difficulty communicating to one another. In our globalized system, language barriers are often faced, especially when conducting multi-party negotiations or in countries with diverse ethnic and linguistic compositions. Some of the issues of inter-party negotiation can be mitigated through the live translation capabilities of NLP (Choyal et al., 2024). Models may have the ability to actively translate both spoken and written word in real time, more quickly than a translator. Thus, it is ensured that all parties can fully participate in the negotiation process. For languages that have deep regional ties, but aren't widely spoken, there may be difficulty locating a translator who can help with negotiation. In such cases, reliance on a machine learning model

could be beneficial to speeding up the process.

4 Public Opinion

An important element of the negotiation process is public opinion surrounding the negotiations. In conflicts where there is higher support for negotiation, the government may feel more pressure to resolve the conflict as such, rather than seeking a military resolution. In a study conducted by Montoya et al., civilians in conflict zones were asked if they supported negotiation as a means to end conflict, alongside a series of potential social, political, and economic predictors. They determined that one’s level of support for vigilante justice and extra-judicial policing were good predictors of negotiation preferences, as well as the level of trust in state institutions and proximity to conflict (Montoya & Tellez, 2020). Interestingly, one’s level of victimization had little predictive power as to their support for negotiations (Montoya & Tellez, 2020). Therefore, cultural indicators may provide the best predictor of public support for negotiation.

5 Limitations and Ethical Concerns

Despite the significant asset that machine learning may prove to offer the field of negotiation, it is still limited in the scope of benefits it can provide. Furthermore, there are ethical considerations that must be taken into account, as with any emerging technology.

5.1 Limitations

While NLP models have significantly improved their logic and reasoning skills, there is still significant room for improvement. Often, NLP models struggle to pass systemic reasoning tests and make ”inconsistent inferences” about the world around them (Molak, 2025). A potential solution to this would be Chain-of-Thought (CoT) prompting, wherein the user

provides an example of how to solve a problem step-by-step or simply prompts the model to do so. It's been found that adding a variation of "think about this step-by-step" leads to the generation of more systematic outputs.

AI models may also experience "hallucinations," wherein they present false information as true. This is of great concern to negotiators, who rely on having accurate information when dealing with parties in contentious situations. The frequency of hallucinations can be reduced through retrieval augmented generation (RAG), which only pulls information from provided documents, rather than internal training knowledge. An example of this would be the models used by Arana-Catania et al., which only used data from the negotiations provided. Yet, the more information that the model has, the better the analysis will be. Limited volume has the potential to skew results and misguide the interpretation of that data (Arana-Catania et al., 2022). Thus for RAG to be an effective solution to hallucinations, it requires significant data to supplement the its lack of outside training data.

5.2 Ethical Concerns

The ethical concerns of AI in negotiation are multifold. Based on what data was used in training, there may be a tendency to favor certain groups (Choyal et al., 2024). As such, it's important to diversify the datasets being used in this capacity, so they mirror the contexts in which they're being used.

Data security is another major concern regarding AI applications in negotiation. The potential exists for privacy to be infringed upon, especially in cases where AI is being used in a monitoring context (Choyal et al., 2024). To solve this, it's important to include privacy measures, such as data anonymization, and consent mechanisms into the analytic methods of ML models.

If in the wrong hands, these models have the potential to spread disinformation or result in targeted attacks of individuals or groups (Choyal et al., 2024). What is yet to happen in the field of AI is to build a strict international framework around its use, like we have with

other emerging technologies that have great potential for both good and bad. Applying an appropriate regulatory framework may even stimulate innovation due to the stability that it would provide (Mariyono & Hidayatullah, 2025).

6 Discussion

ML has the opportunity to be incredibly beneficial to negotiation, especially in the era of protracted, intrastate conflict which we have today. There are large datasets on which to train negotiation models and due to their regional concentrations, any supplemental data provided can be well focused. The translation capabilities of NLP would be useful to allow for more effective dialogue between mediators and parties. However, there are still immense limitations and ethical concerns that need to be addressed before AI becomes more widespread in its use. Hallucination is concerning, especially when acting under the assumption that people tend to over-extend their trust to these tools. Additionally, without a cohesive legal framework under which countries are allowed to use ML, these tools will likely end up being used beyond their intended purposes. It's important that there is a structure in place to regulate AI tools before they are used widespread, so it is clear who is accountable for the decisions made by the model, especially if they end up misstepping.

7 Conclusion

There is a lot of fear being expressed about the integration of AI in conflict resolution, just as much as there is interest in its use. However, one thing is clear: our traditional operational methods are changing and it's important to be able to adapt to them. Further work needs to be done to streamline our models for negotiations, resolving some of the outstanding issues they face and ethical concerns need to be addressed. If that happens, there is potential for a great asset to be provided to negotiation methods.

References

- Arana-Catania, M., van Lier, F.-A., & Procter, R. (2022). Supporting peace negotiations in the yemen war through machine learning. *Cambridge University Press*.
- Aydoğar, R., Baarslag, T., & Gerding, E. (2021). Artificial intelligence techniques for conflict resolution. *Group Decision and Negotiation*.
- Choyal, P., Mishra, P., Sharma, J., & Chaturvedi, A. (2024). Navigating peace: Harnessing artificial intelligence for conflict resolution. *International Research Journal of Modernization in Engineering Technology and Science*.
- Mariyono, D., & Hidayatullah, A. A. (2025). Quantifying ai's role in conflict resolution: Assessing its potency in mediation and peacekeeping. *Dirasat: Human Social Sciences*.
- Molak, A. (2025). Ai in conflict resolution: Practical considerations, opportunities and challenges. *Conflict Resolution Quarterly*.
- Montoya, A. M., & Tellez, J. (2020). Who wants peace? predicting civilian preferences in conflict negotiations. *Journal of Politics in Latin America*.
- Pauwels, E. (2020). Artificial intelligence and data capture technologies in violence and conflict prevention: Opportunities and challenges for the international community. *Global Center on Cooperative Security*.